575 Lab

Opening the Black Box

Mechanistic Interpretability for AI Agent Tool Selection Using Sparse Autoencoders

Hannes Hapke with **David Cardozo** · 575 Lab, Dataiku Inc.

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ABOUT THE SPEAKER

Hi, I'm Hannes Hapke

Director, Engineering at Dataiku, 575 Lab
Google Developer Expert -- Machine Learning

Focus areas:

- Open Source Machine Learning
- Production ML systems & pipelines
- Generative AI and agentic systems

Co-author of four O'Reilly & Manning books, including:

- **Generative AI Design Patterns** (2025)
- **Machine Learning Production Systems** (2024)

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ABOUT THE LAB

Dataiku's Open Source Office

Advancing Responsible AI through open source — production-ready tools that give enterprises **transparency, control, and governance** to deploy AI with confidence.

Our Responsible AI work:

- **Trust** — no black boxes, no hidden behaviour
- **Control** — no vendor lock-in, no opaque dependencies
- **Accountability** — auditable, reproducible, community-reviewed

SIGNATURE PROJECTS

- **Kiji Inspector** — mechanistic interpretability for agent tool selection *(this talk)*
- **Kiji Privacy Proxy** — detect & redact sensitive data before LLM API calls

ALSO CONTRIBUTING TO

vLLM · scikit-learn (sponsor) · Cardinal (active learning) · CodeMirror

dataiku.com/open-source

The Problem: Opaque Agent Decisions

AI agents autonomously select tools (databases, web search, code execution, ...)
based on natural language requests.
But **why**?

> *"Find information about our company's API rate limits."*
> *→ Internal docs search? Or public web search?*

Current approaches **fail** to provide mechanistic insight:

1. **Prompt engineering** -- reveals correlations, not causal mechanisms
2. **Behavioral testing** -- characterizes inputs/outputs, not internals
3. **Chain-of-thought** -- plausible narratives $\neq$ true computation

> We need to look **inside the model**.

PART I

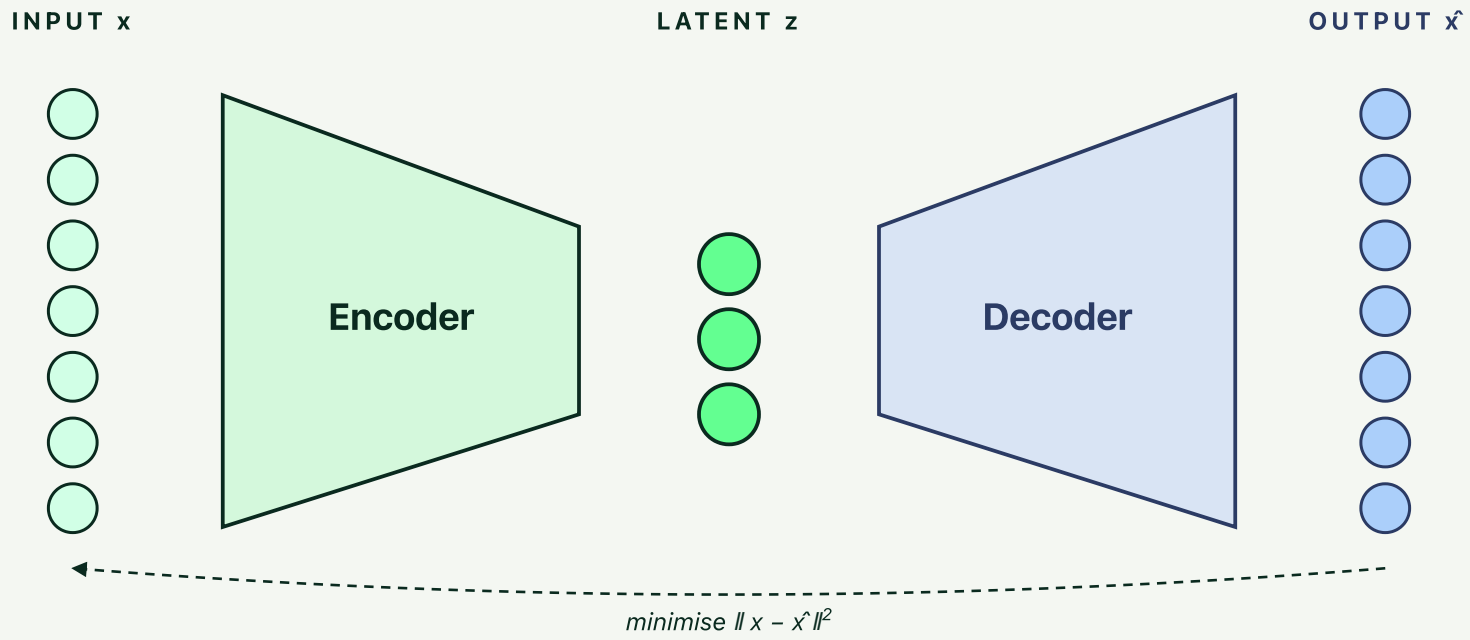
The Solution

LOOKING INSIDE WITH SPARSE AUTOENCODERS

How to Understand Models — Step 1

AUTOENCODERS

A **self-supervised** technique that learns new representations: compress an input through a bottleneck, then reconstruct it. What survives the bottleneck is what the model considers essential.

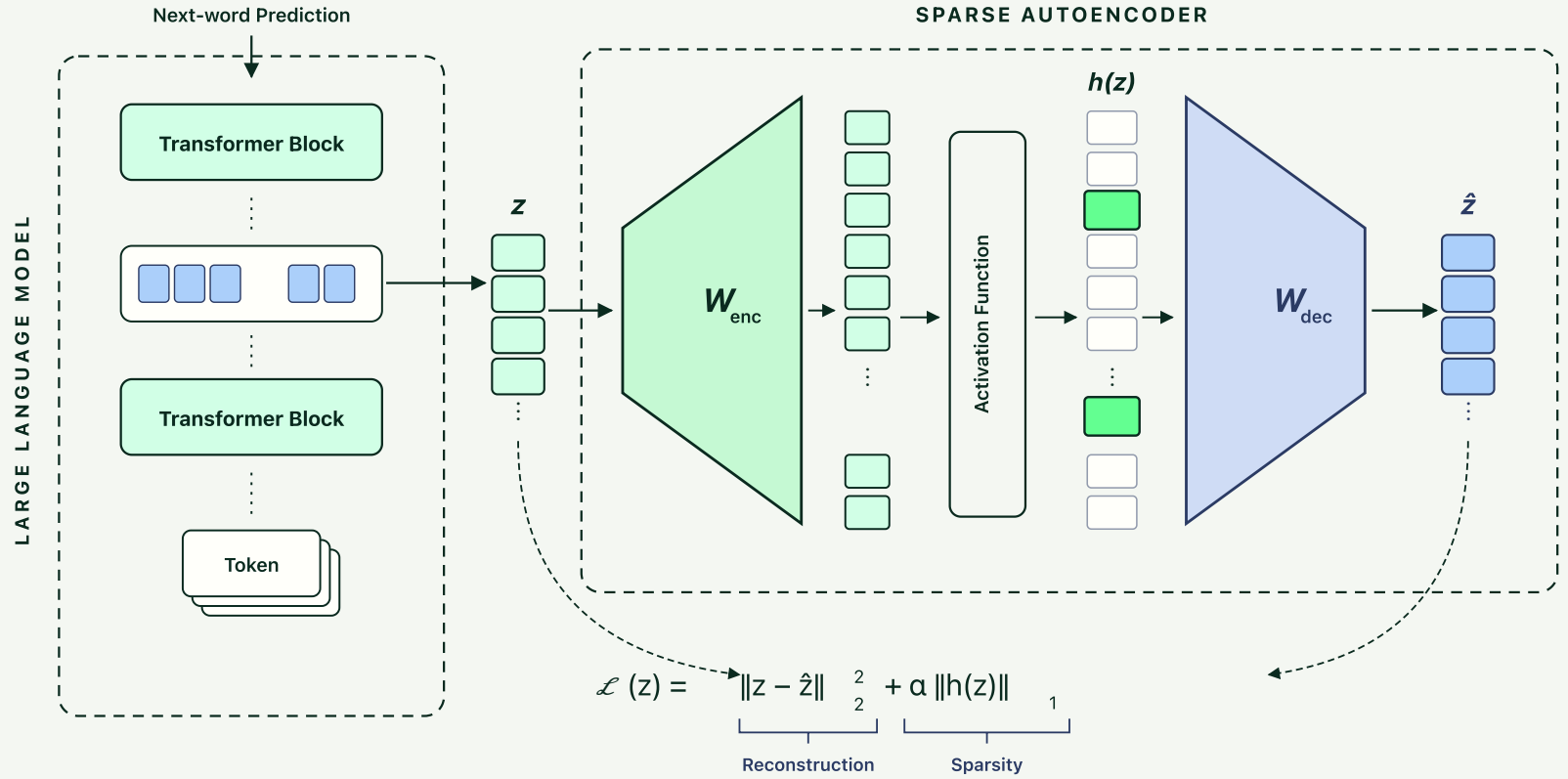


- **Well-understood** -- decades of theory and practice
- **Lossy by design** -- the bottleneck forces the model to *prioritise*

How to Understand Models — Step 2

SPARSE AUTOENCODERS (SAE)

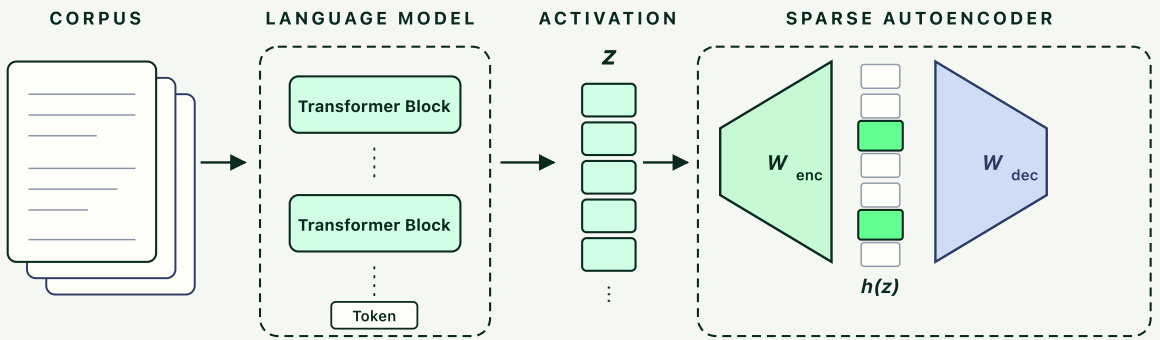
Flip the autoencoder on its head: instead of compressing, **expand** the latent space — but force fewer than **5%** of dimensions to fire on any given input.



How to Understand Models — Step 3

INTERPRETING WHAT THE FEATURES MEAN

A trained feature is just an index. To *label* it, collect the contexts where it fires most strongly — then let an LLM describe the pattern.



for each feature i, collect top-activating contexts

FEATURE

$h(z)_i$

← i →

FEATURE DICTIONARY

#	Top-activating contexts	Label
#12	`"import numpy as np…"`	Python imports
#47	`"GET /api/v2/orders…"`	REST endpoints
#88	`"rate limit exceeded…"`	Throttling errors
#103	`"our company's internal…"`	Internal scope
…	…	…

INTERPRETABLE THEMES

- ⊕ Programming-language syntax
- ⊕ Web / API vocabulary
- ⊕ Internal-vs-external scope
- ⊕ Error / failure-mode language

The Kiji Inspector

01

MODEL AGNOSTIC

Framework to train custom Sparse Autoencoders — works on **any open-source LLM**, not just the ones we tested.

02

END-TO-END TRAINING PIPELINE

No custom training setup needed. The project **generates the contrastive datasets** for you and runs the full extract → train → analyse loop.

03

PRODUCTION INFERENCE SUPPORT

vLLM patches, single-endpoint Docker containers, and PyPI packages — ready to drop into a live serving stack.

04

FULLY OPEN SOURCE

Pre-trained SAE models distributed via **Hugging Face**, **Docker Hub**, and **PyPI** — reuse without retraining.

GitHub: github.com/dataiku/kiji-inspector

What's Novel?

01

DECISION-TOKEN EXTRACTION

Capture activations at the *precise moment* of tool commitment — not averaged over the prompt.

02

PAIRS AS POST-HOC PROBES

The SAE learns the model's natural vocabulary **unsupervised**. Contrastive pairs are statistical probes only — never training signal.

03

TOKEN-LEVEL FUZZING EVALUATION

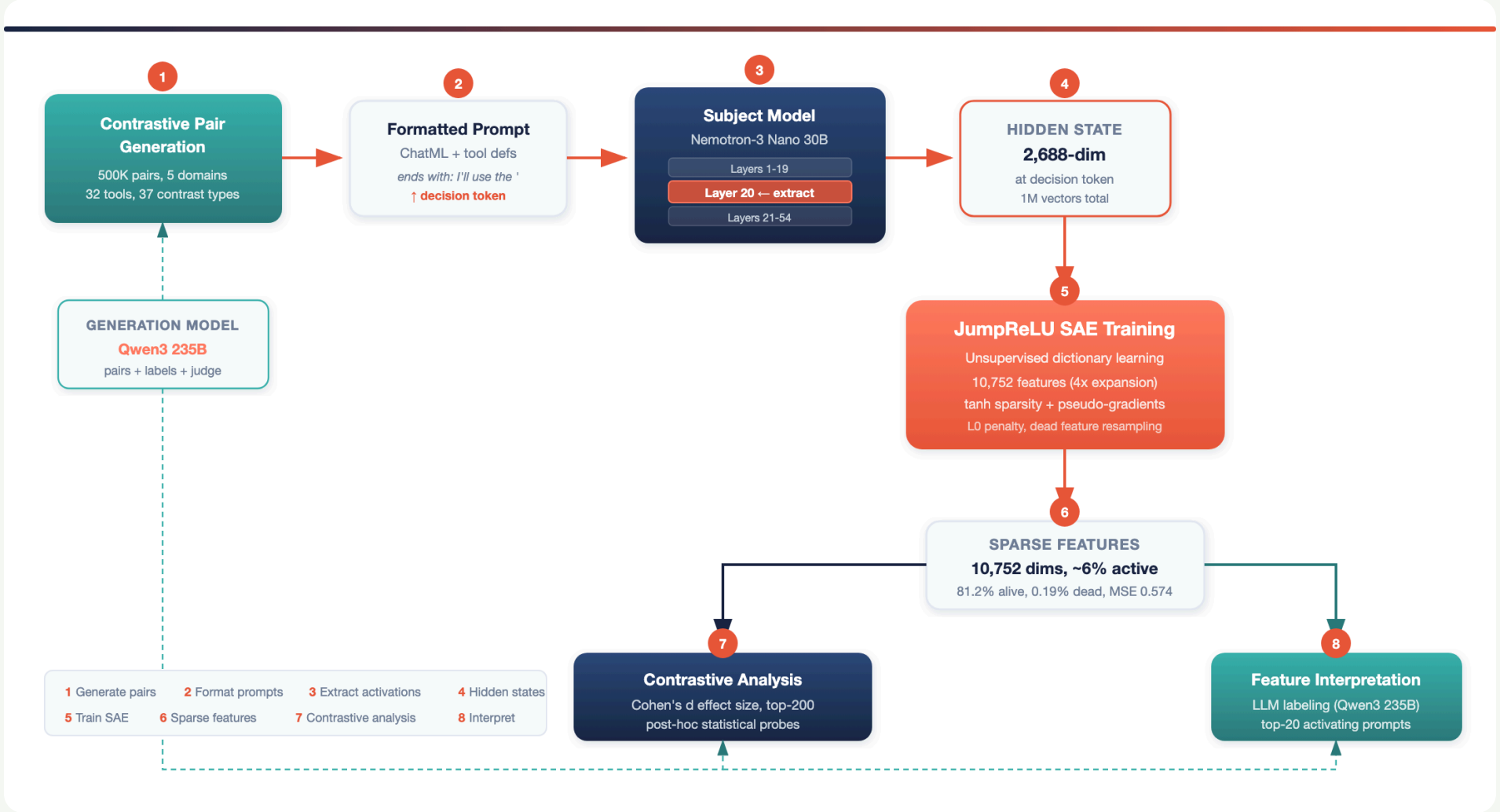
Adapted from Eleuther AI's autointerp — catches labels that are *"right for the wrong reasons."*

04

CAUSAL VALIDATION VIA ABLATION

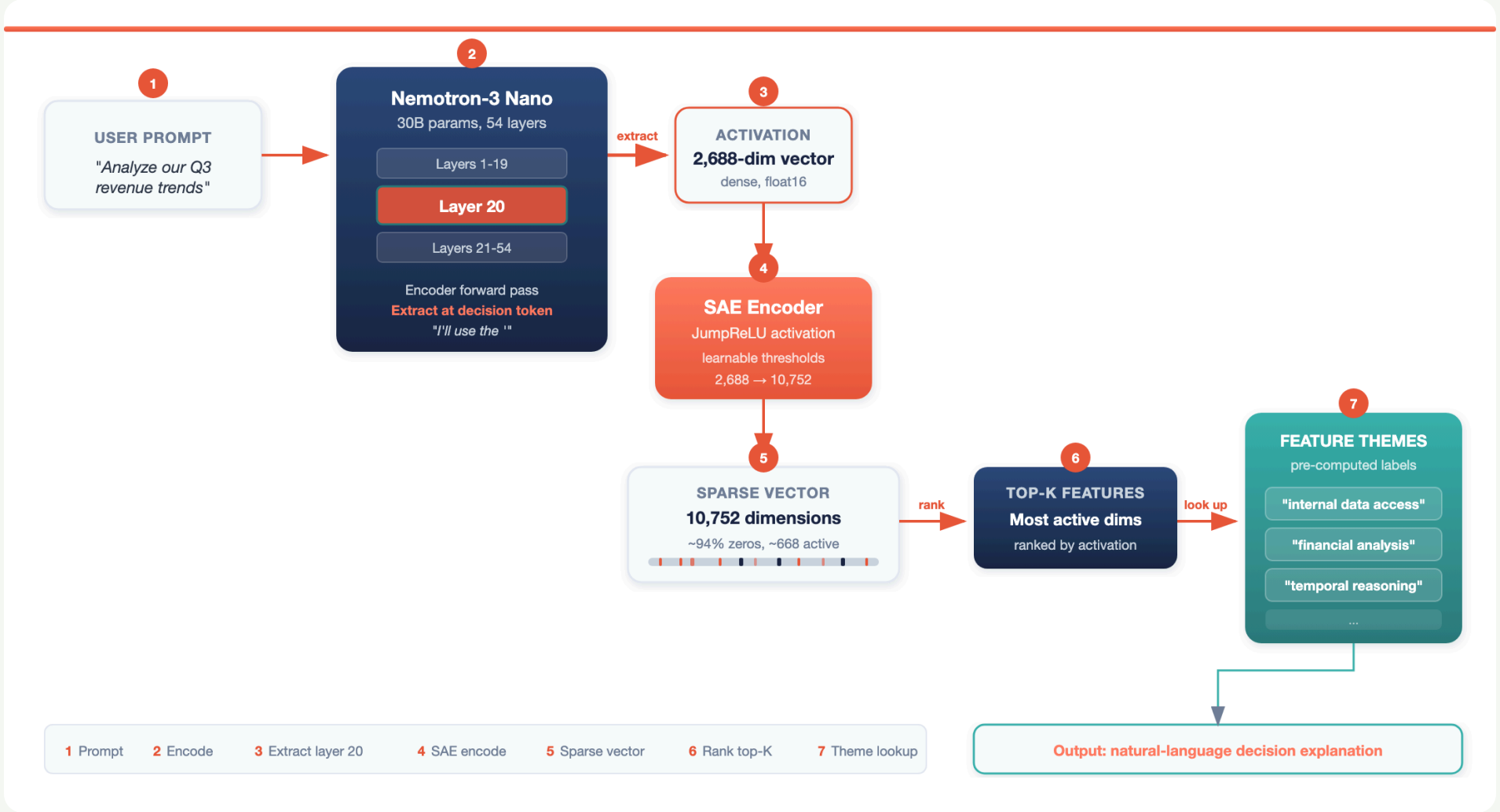
Zero a feature, measure the prediction flip — turns correlations into evidence.

Our Complete Training Pipeline



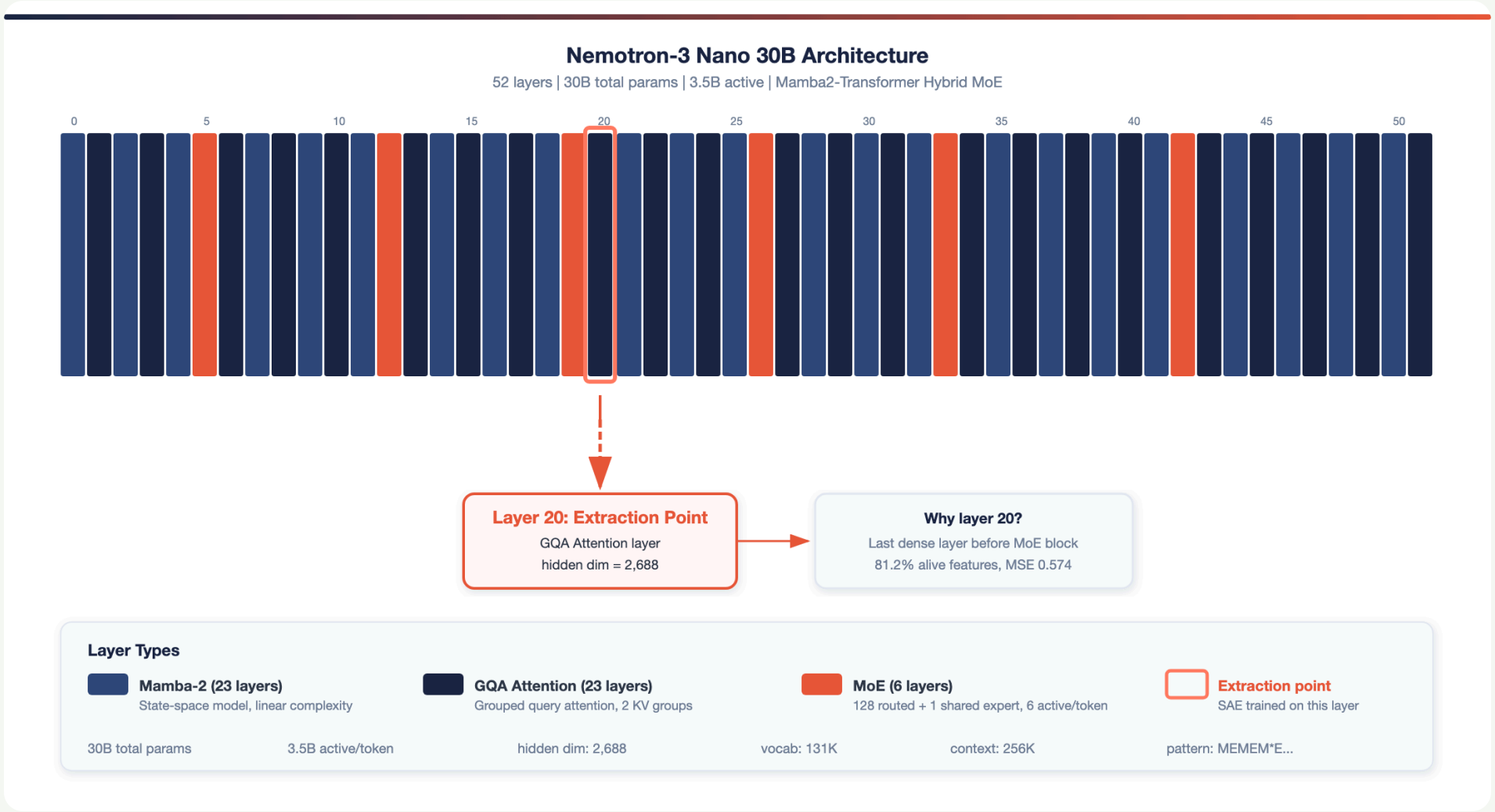
Contrastive pairs are generated and encoded by the subject model. The SAE is trained unsupervised on the extracted

Our Inference Setup



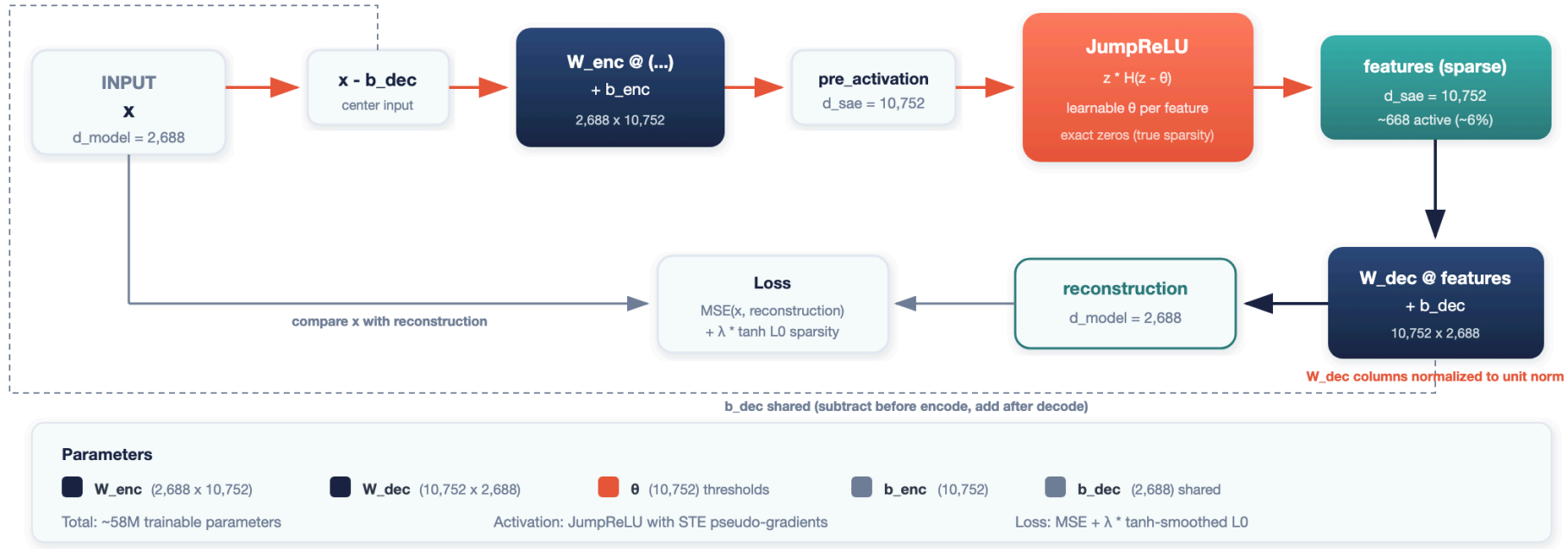
Seven steps from raw prompts to human-readable decision explanations.

Nemotron-3 Nano Architecture



Hybrid Mamba2-Transformer MoE, 52 layers, open weights — chosen for routing diversity (MoE) and a tractable extraction

PyTorch SAE Model Architecture



Encoder projects 4,096-dim input to 16,384 sparse features via JumpReLU with learnable per-feature thresholds. Decoder

Decision Token Extraction

Every formatted prompt ends with:

```
<|assistant|> I'll use the '
```

The hidden state at this final token is the **decision token** -- the model's internal state at the moment it commits to a tool name.

- Activations extracted at **layer 20** of Nemotron-3-Nano-30B (54-layer MoE) — *layer choice justified below*
- Hidden dimension: **4,096**
- Batched extraction with left-padding for alignment
- Dataset: **1,000,000** activation vectors (500K contrastive pairs)

Hands-on: Extract a Layer's Activations

The Kiji Inspector builds on standard PyTorch primitives. Here's the extraction step in raw `transformers` — the same mechanism, no Kiji magic, on any HuggingFace causal LM.

```
pip install -U -q kiji-inspector transformers
```

Setup — Imports & Config

```
import torch
from transformers import AutoModelForCausalLM, AutoProcessor

LAYER_INDEX = 8
MODEL_ID    = "google/gemma-4-E4B-it"
PROMPT      = "My dishwasher is smelly, what is the first element I should review?"
```

Pick the **layer** to study and the **model** to study it on. Everything else flows from these three constants.

Load Model & Processor

```
processor = AutoProcessor.from_pretrained(MODEL_ID)
model = AutoModelForCausalLM.from_pretrained(
    MODEL_ID,
    torch_dtype=torch.bfloat16,
    device_map="auto",
)
model.eval()
```

`bfloat16` halves memory; `device_map="auto"` lets HF Accelerate place layers across GPUs. `.eval()` disables dropout — we want deterministic activations.

Format the Prompt

```
messages = [{"role": "user", "content": [{"type": "text", "text": PROMPT}]}]
prompt = processor.apply_chat_template(
    messages, tokenize=False, add_generation_prompt=True,
)
inputs = processor(text=prompt, return_tensors="pt")
inputs = {k: v.to(next(model.parameters()).device) for k, v in inputs.items()}
```

`add_generation_prompt=True` appends the assistant turn header — the model is now poised to *answer*. That final position is the **decision token** we just defined.

Capture Activations with a Forward Hook

```
captured = {}

def hook(_module, _inputs, output):
    hidden = output[0] if isinstance(output, tuple) else output
    captured["tensor"] = hidden.detach().cpu()

layer = model.model.language_model.layers[LAYER_INDEX]
handle = layer.register_forward_hook(hook)
```

PyTorch's `register_forward_hook` intercepts the layer's output mid-forward-pass. We `.detach()` to drop the autograd graph and `.cpu()` so we don't pin GPU memory.

Run Inference, Retrieve the Tensor

```
try:
    with torch.inference_mode():
        model(**inputs)
finally:
    handle.remove()

hidden_state = captured["tensor"]
```

`inference_mode()` is faster than `no_grad()` — it also skips view-tracking. The `try / finally` guarantees the hook is removed even on error. `hidden_state` is ready to feed into a trained SAE.

Now: Ask the Kiji Inspector What Fired

```python
from kiji_inspector import SAE

sae, feature_descriptions = SAE.from_pretrained(
    base_model="google/gemma-4-E4B-it",
    layer=8,
)

# Activation for the first sequence, last token
last_token_act = hidden_state[0, -1, :]

# Describe the top features activating on this token
sae.describe(last_token_act, feature_descriptions)
```

`SAE.from_pretrained` pulls a layer-matched SAE *and* its labelled feature dictionary from the Hub. `describe()` returns the highest-activating features and their human-readable labels — the decision token, explained.

Output: Top Features on the Token

```
[(2341, 'unknown', 7.57),
 (14641,
  {'label':           'Dishwasher Gasket Issues',
   'description':     'Detects descriptions of dishwashers leaking from '
                      'the door or bottom, often mentioning a worn, torn, '
                      'or cracked door gasket.',
   'confidence':      'high',
   'mean_activation': 7.66,
   'max_activation':  8.63,
   'frac_nonzero':    1.0,
   'top_examples':    ['My 18-yr-old dishwasher is leaking from the door...',
                       ...],
   'bottom_examples': ['Find general best practices for API versioning...',
                       ...]},
  7.19),
 ...]
```

A labelled feature (**#14641**) semantically matches the prompt — with full provenance (confidence, activation stats, witnessing contexts). Feature **#2341** fired too but autointerp hasn't named it. The trailing float in each tuple is the

Contrastive Pair Design

Pairs share the same *intent* but require *different tools*:

Shared Intent	Anchor (tool A)	Contrast (tool B)
Resolve password issue	"How do I reset my password?" → `knowledge_base`	"I tried resetting 3 times but the email never arrives" → `ticket_lookup`
Evaluate energy stocks	"Which companies invest in renewables?" → `financial_analysis`	"Which stocks trade below book value?" → `market_data_lookup`
Check product version	"What is the latest version?" → `file_read`	"Set the version to v3.2.1" → `file_write`

5 domains, 32 tools, 37 contrast types.

JumpReLU SAE Architecture

JumpReLU = ReLU with a learnable per-feature threshold — gives exact zeros but stays trainable via tanh pseudo-gradients (Rajamanoharan et al., 2024).

Encoder:

$$f_i(\mathbf{x}) = \pi_i(\mathbf{x}) \cdot H(\pi_i(\mathbf{x}) - \theta_i)$$

where $\pi_i(\mathbf{x}) = [W_{\text{enc}}(\mathbf{x} - \mathbf{b}_{\text{dec}}) + \mathbf{b}_{\text{enc}}]_i$ and θ_i is a learnable threshold.

Key properties:

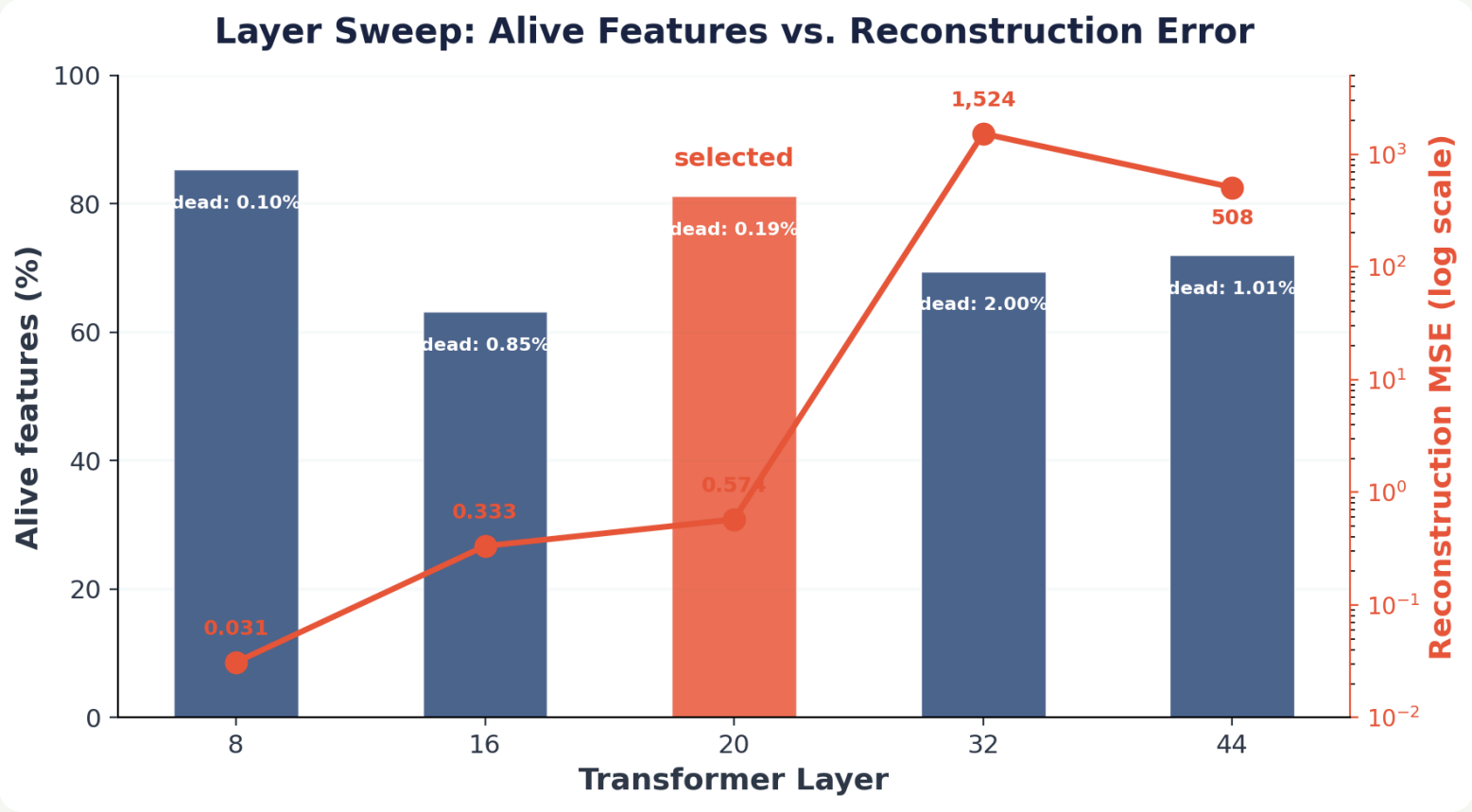
- **Exact sparsity** -- Heaviside step function H produces true zeros
- **Smooth training** -- tanh approximation for gradient flow:

$$\hat{\mathcal{L}}_{\text{sparse}} = \sum_{i=1}^{M} \text{ReLU}\left(\tanh\left(\frac{\pi_i - \theta_i}{\varepsilon}\right)\right)$$

- **Pseudo-gradients** via rectangular kernel density estimator

Dictionary size $M = 16{,}384$ ($4\times$ hidden dim).

Why Layer 20?



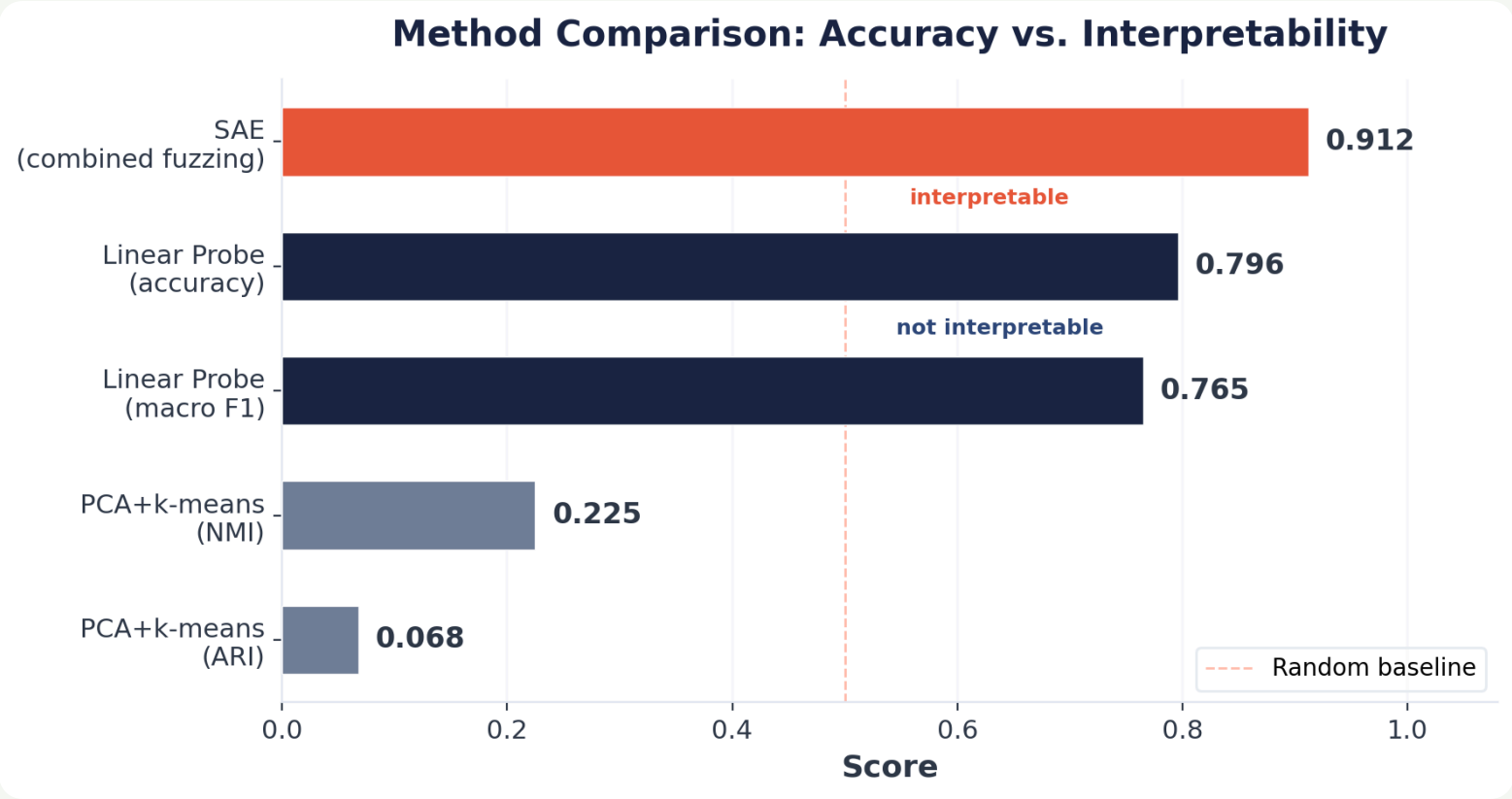
- Layers 8/16: low MSE but *pre-decision* representations
- **Layer 20**: best alive %, lowest dead %, MSE < 1.0
- Layers 32+: MoE expert routing → 500x+ higher MSE

SAE Feature Health (Layer 20, Full Dataset)

Metric	Value
Total features	16,384
Alive features (>0.1% firing)	**81.2%** [80.6, 81.8]
Dead features (0% firing)	**0.19%** [0.13, 0.27]
L0 (active features per input)	668 (≈4.1% density)
Reconstruction MSE	0.574

The SAE efficiently uses its capacity: sparse encoding with high feature utilization.

Baselines: Why Not Just a Probe?



- Linear probe confirms tool identity is *linearly encoded* (79.6% across 32 classes) -- but provides *no interpretability*
- PCA + k-means fails entirely -- tool signal is not dominant variance
- The SAE bridges this gap: **interpretable** *and* **causally testable** features

Token-Level Fuzzing Evaluation

Autointerp (EleutherAI) is the standard recipe for labelling SAE features at scale: an LLM proposes a label from top-activating contexts, a second LLM judges whether the label predicts activation. We tighten the test.

Standard evaluation: "Does this label predict which *prompts* activate the feature?"
Our evaluation: "Does this label predict which *tokens* activate the feature?"

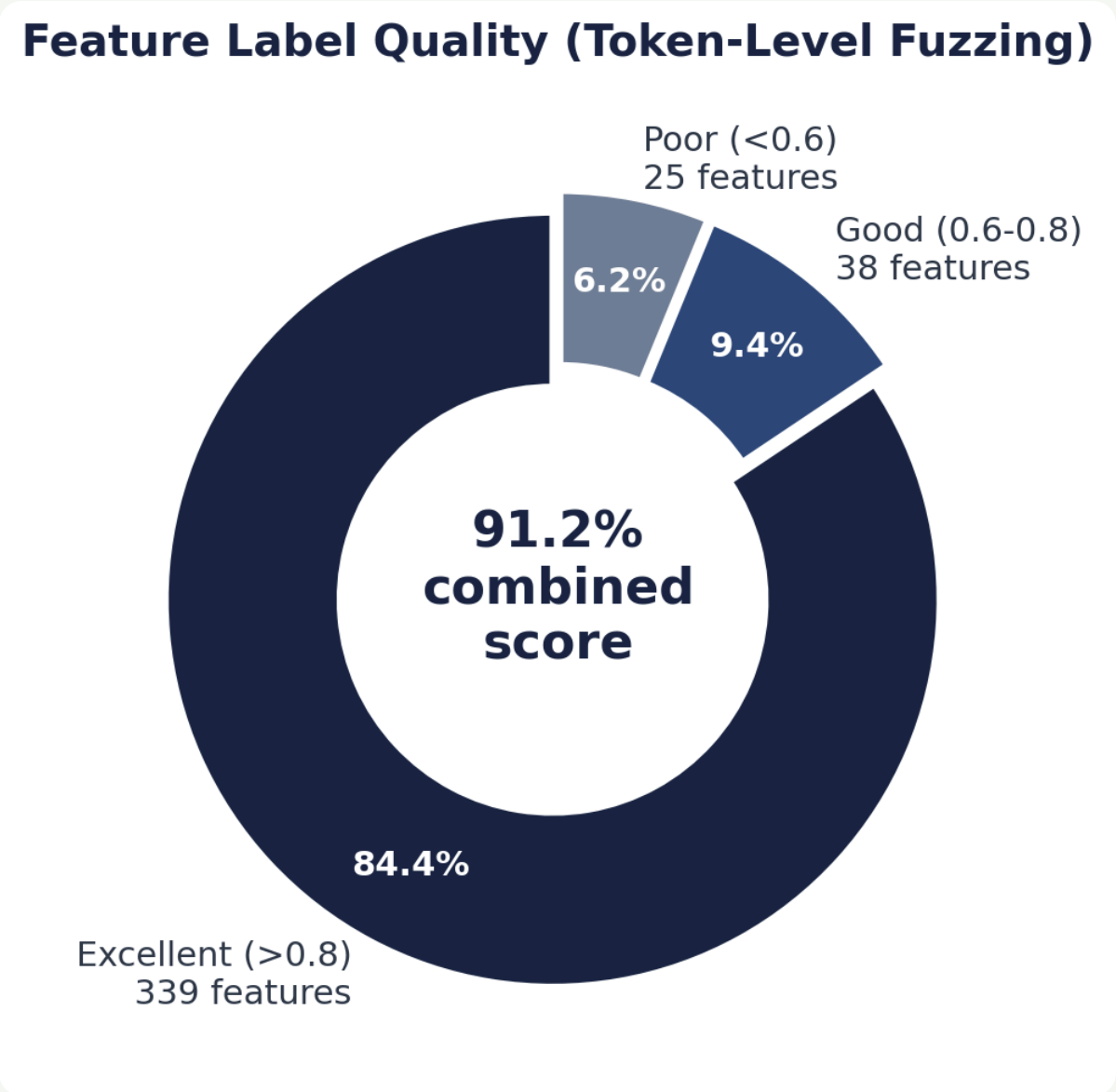
Protocol:

1. Extract per-token activations across entire prompt
2. Highlight top-K tokens in user request span
3. A/B test: LLM judge picks which highlighted text matches the label
4. Randomized order to prevent position bias

Combined score = $0.7 \cdot \text{acc}_{\text{token}} + 0.3 \cdot \text{acc}_{\text{prompt}}$

Token-level gets higher weight because it tests the *actual mechanism*.

Fuzzing Results: Features Are Interpretable



PART II

The Causality Test

From Correlation to Causal Evidence

Feature Ablation: Experimental Design

Question: Are contrastive features *causally necessary* for tool selection, or merely correlated?

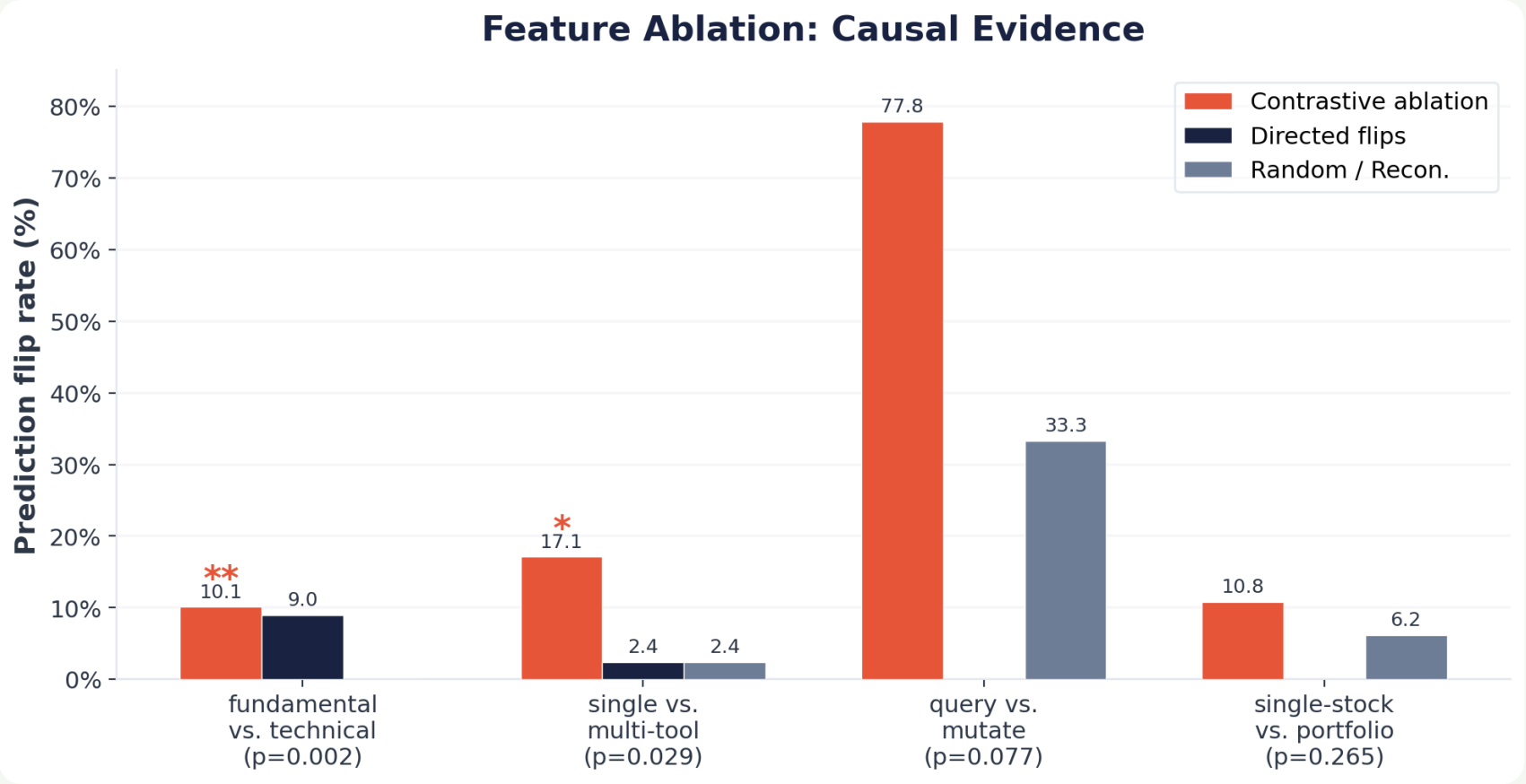
Method:

1. Intercept residual stream at layer 20
2. Encode through trained SAE
3. **Zero out top-10 contrastive features**
4. Decode back into residual stream
5. Measure: does the model's tool prediction *flip*?

Controls:

- **Random ablation**: zero 10 random non-contrastive features
- **Reconstruction-only**: SAE encode → decode with *no* features zeroed (measures round-trip distortion)

Ablation Results: Causal Evidence



Aggregate (23 types): 16.1% contrastive vs. 13.0% reconstruction-only

Why This Matters: Interpreting the Ablation

Fundamental vs. Technical Analysis ($p = 0.002$):

- Zeroing 10 contrastive features flips **10.1%** of predictions
- 9.0% flip *toward the contrast tool* (directed change)
- Random ablation: **0%** flips. Reconstruction-only: **0%** flips
- These features are *causally necessary* for this distinction

Critical control insight:

> *Random ablation flip rate equals SAE round-trip distortion across all 23 contrast types. Removing 10 random features from ~668 active adds no disruption beyond the encode-decode cycle itself.*

This validates the design: ablation effects are due to *specific features*, not general signal degradation.

The Spectrum of Causal Involvement

Not all decisions rely on sparse feature circuits. Across 23 contrast types we see a spectrum:

- **Sparse circuits** -- a small minority (e.g. fundamental/technical, single/multi-tool) concentrate causal signal in a handful of identifiable features; ablating 10 contrastive features flips up to 10.1% of predictions
- **Distributed encodings** -- many contrast types (e.g. preventive/reactive maintenance) show 0% flips even with 10 features ablated, robust to any 10-feature subset
- **Intermediate** -- the remainder show detectable but not statistically dominant feature involvement

This reveals a heterogeneous landscape:

> *Some tool-selection decisions are governed by interpretable sparse circuits; others rely on distributed, redundant encodings.*

Both findings are scientifically valuable.

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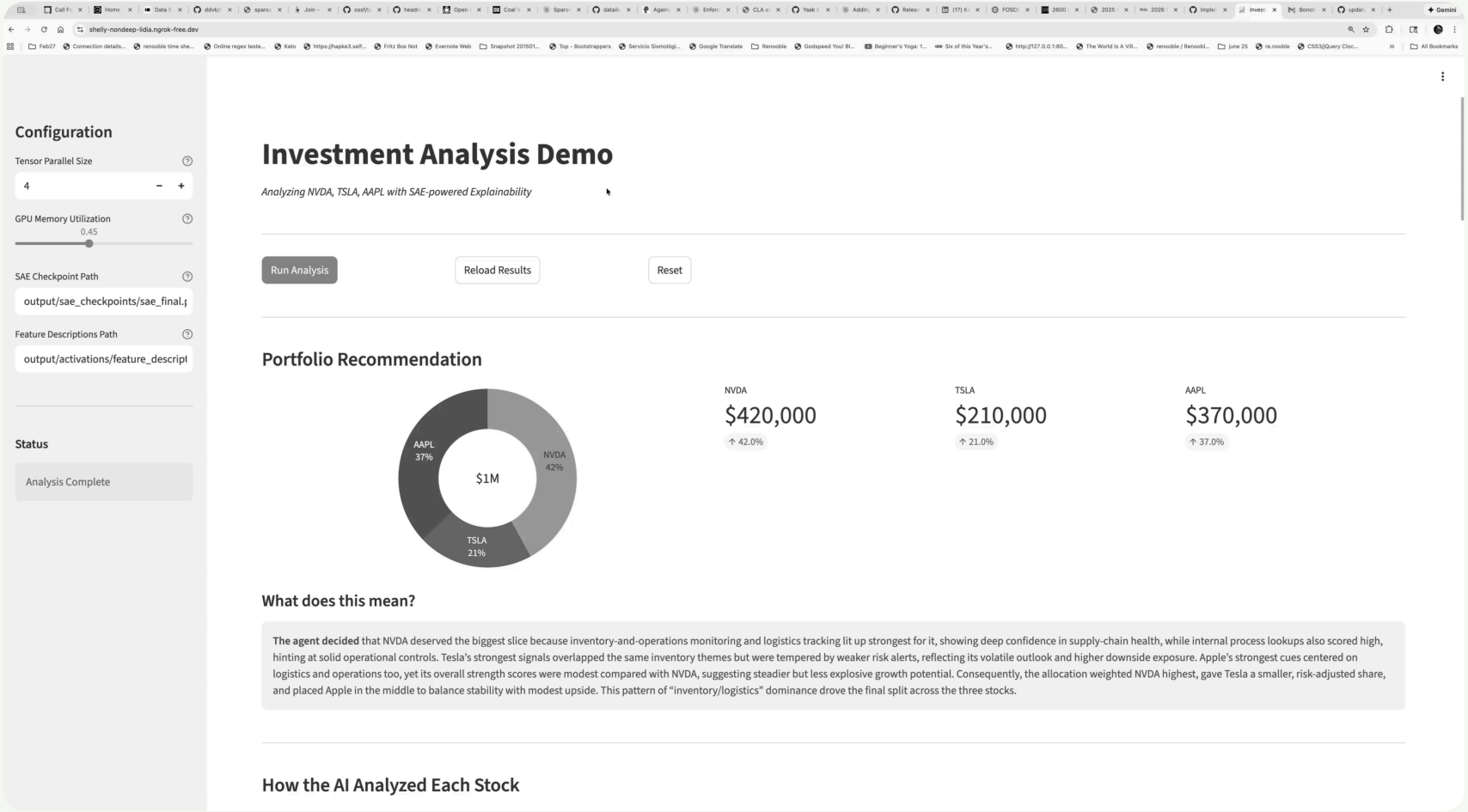
Demo Time

KIJI INSPECTOR, LIVE

End-to-End Demo Application



Interactive system surfacing SAE-derived explanations alongside agent output -- translating internal feature activations into natural-language rationales.

Key Takeaways

01

SAEs discover interpretable decision features without supervision — 91.2% fuzzing accuracy

02

Token-level fuzzing catches labels *"right for the wrong reasons"* — a stricter test than prompt-level evaluation

03

Causal evidence via ablation: specific features are necessary for specific tool-selection decisions ($p = 0.002$)

04

The reconstruction-only baseline is essential — it separates genuine causal effects from SAE distortion artifacts

Limitations and Future Directions

CURRENT LIMITATIONS

- Training is compute-intensive (235B generation model + 30B subject model)
- Access to model is required
- Synthetic contrastive pairs may miss real-world decision factors

FUTURE WORK

- Support for more open-source models like Qwen
- Multi-layer / circuit-level analysis
- Cross-model transfer (do tool-selection circuits generalize?)

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Thank You

Open source: github.com/dataiku/kiji-inspector

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